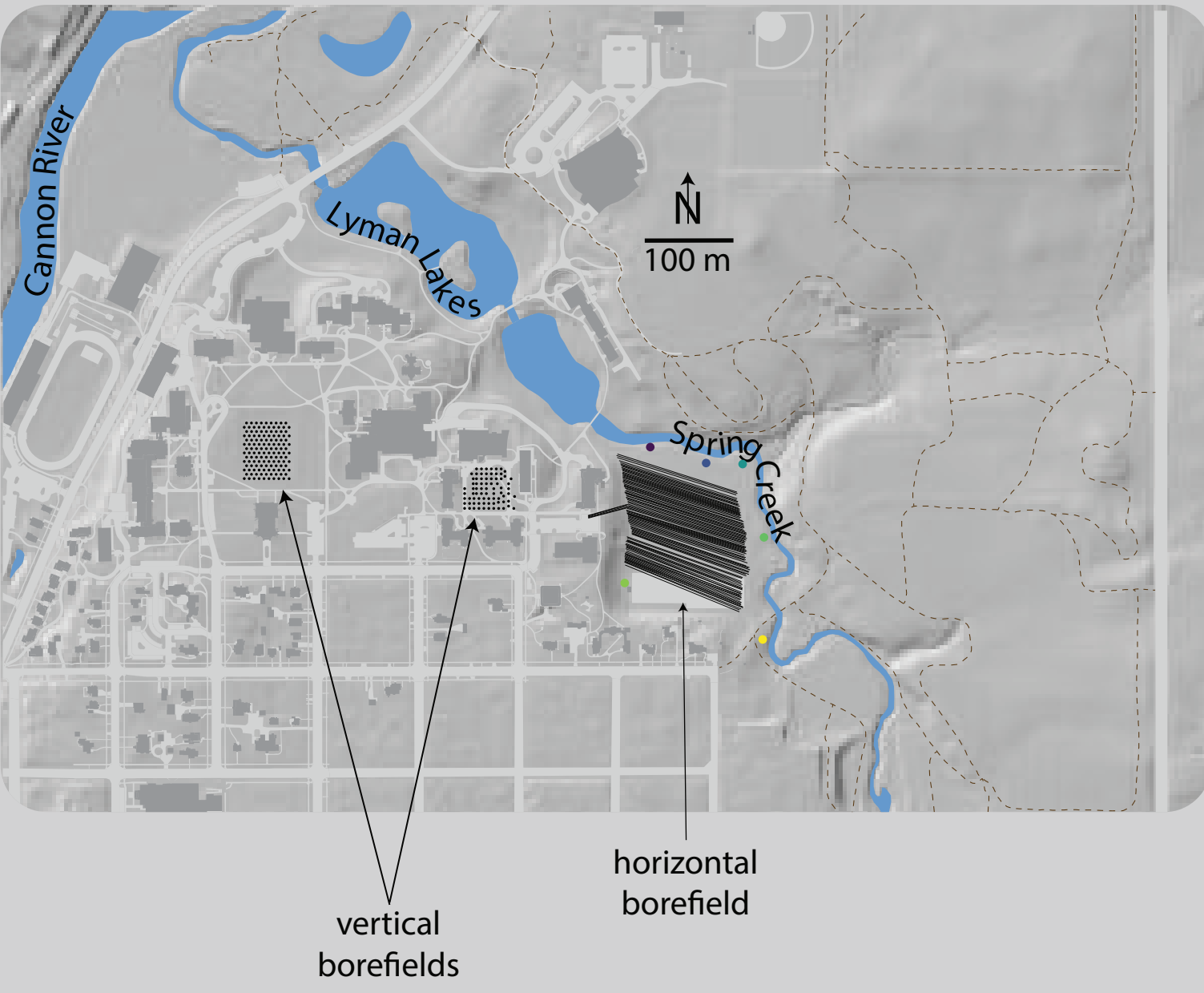


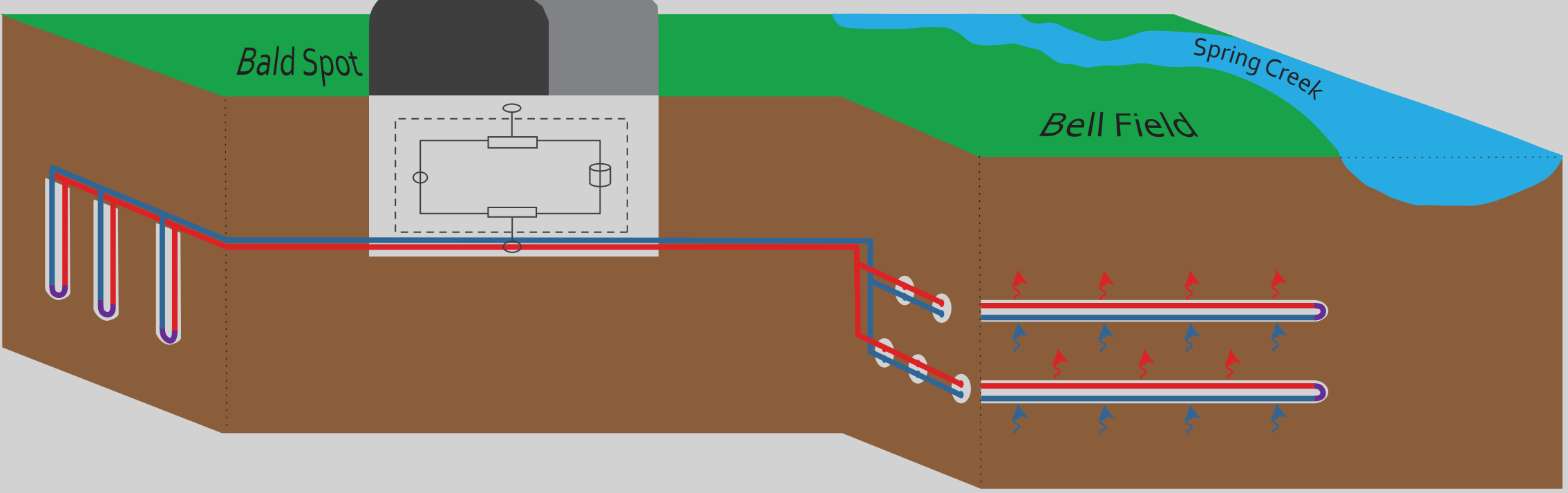
Horizontal geothermal systems at Carleton College

Eve Mullen, Soren Miner, Jasper Priest

Carleton's geothermal system



In 2018, Carleton installed a district-scale closed-loop geothermal heat exchange system to heat and cool campus buildings. It's powered by three borefields, one of which consists of shallow, horizontal bores in unconsolidated sediment (Bell Field).



The Bell Field boreholes alternate between 9 and 14 meters below the land surface.

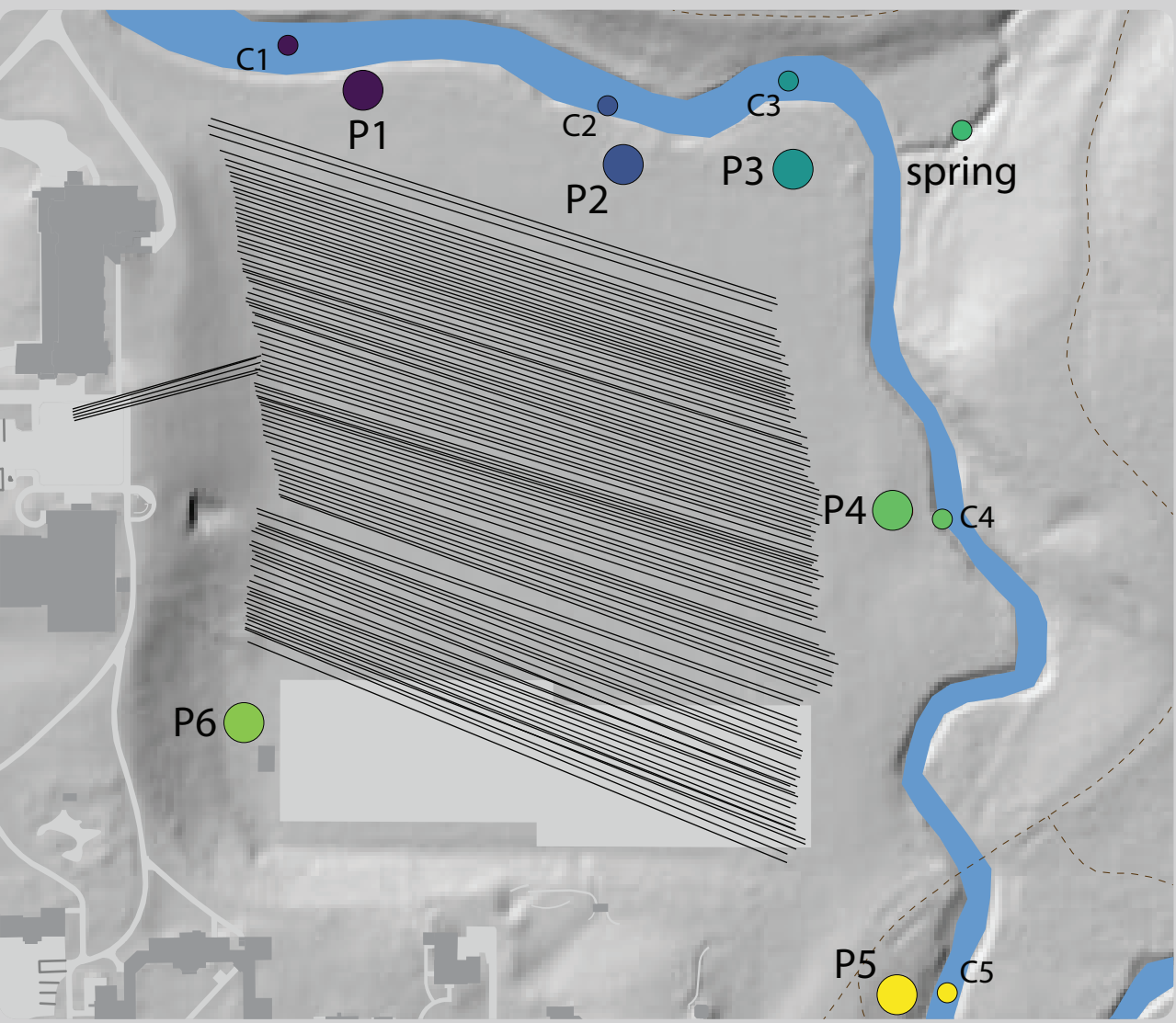
Hydrogeologic context for Bell Field



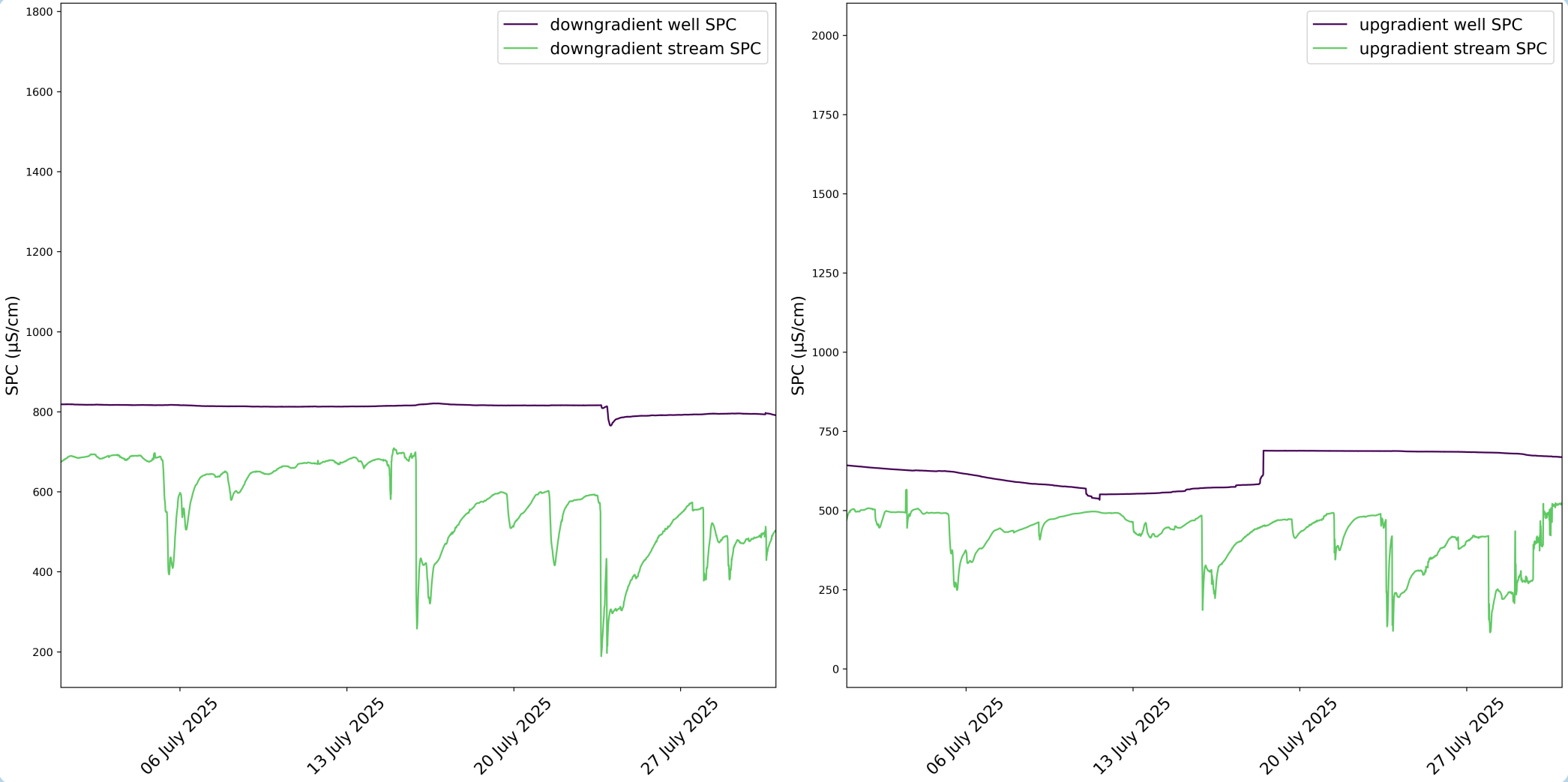
Bell Field is bordered on its north and east by Spring Creek, a small groundwater-fed stream. Northfield is built on top of a layer of partially dissolved limestone (karst), which facilitates fast groundwater flow through the upper layer of bedrock. The Bell Field boreholes are at least 3 meters above this layer.

Monitoring

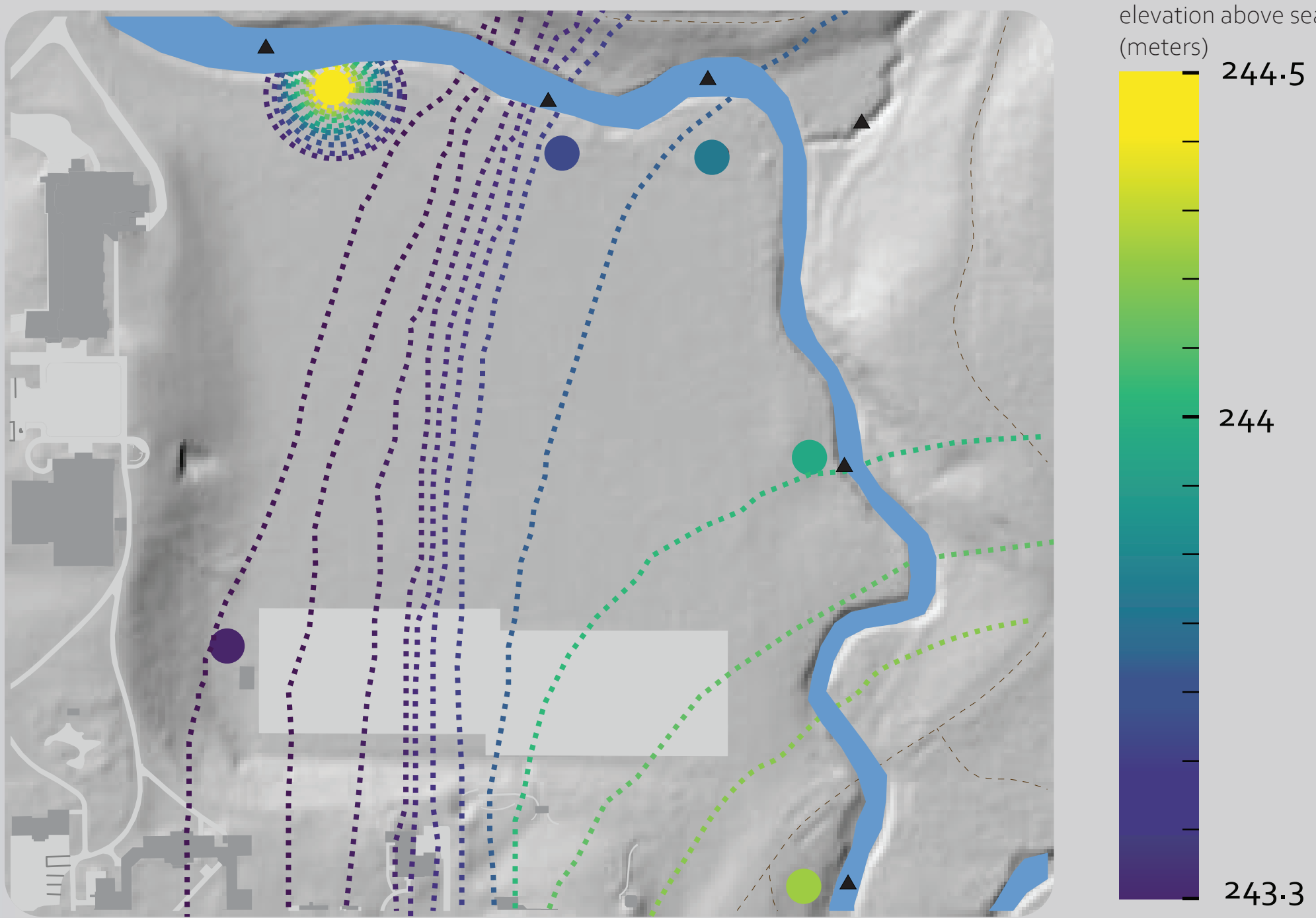
We installed monitoring wells along the edge of Bell Field, where we record temperature and dissolved solids to look for any thermal impacts of the geothermal system on the shallow groundwater.



How do we know the monitoring wells can measure the borefield?



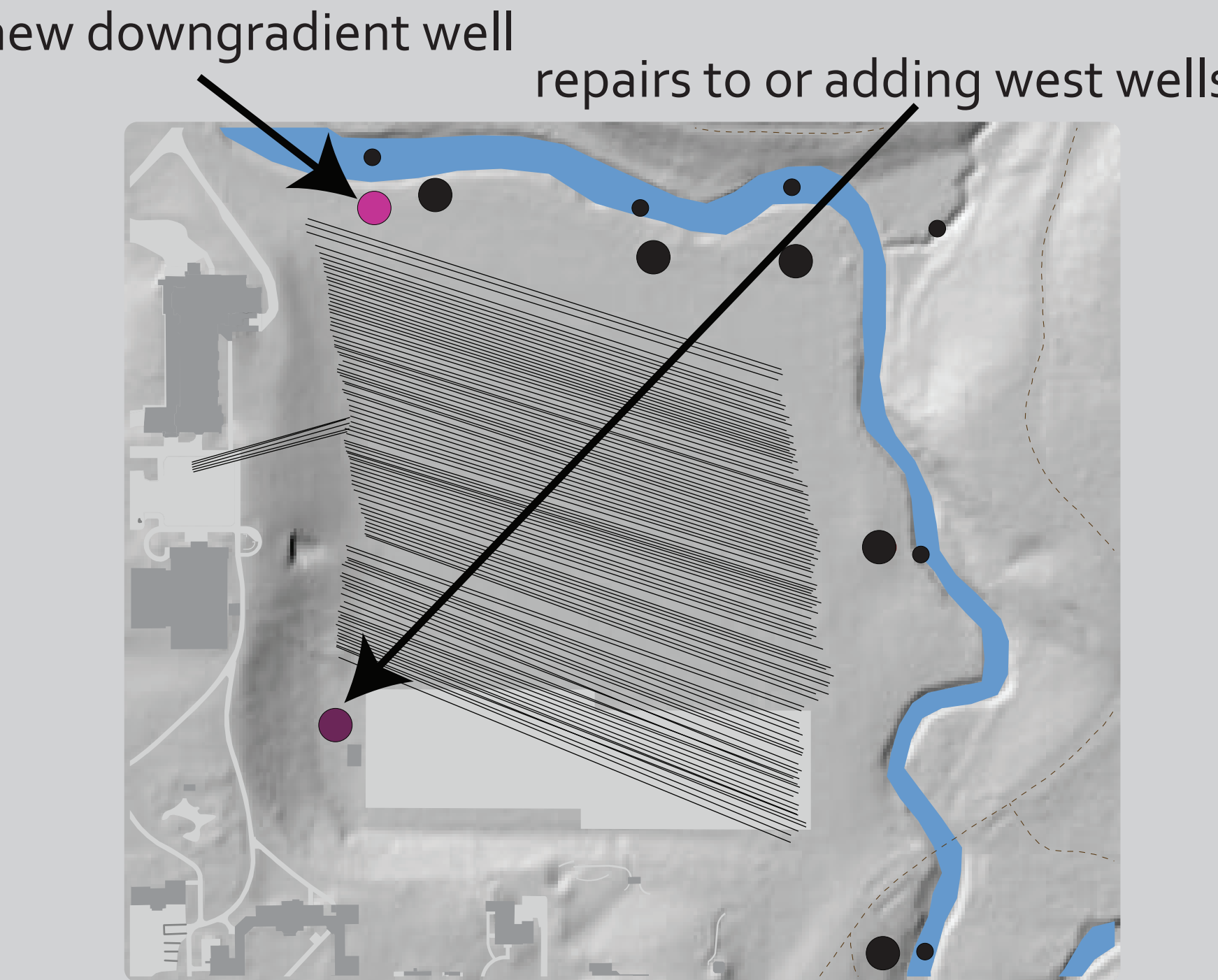
Dissolved solids are significantly less stable in the creek than in the monitoring wells, indicating that the wells reflect deeper groundwater, not surface runoff percolating to the shallow water table.



Using water table elevation, water chemistry, and the known regional groundwater flow direction, we interpolate a groundwater surface and estimate a flow direction through the borefield.

Future monitoring

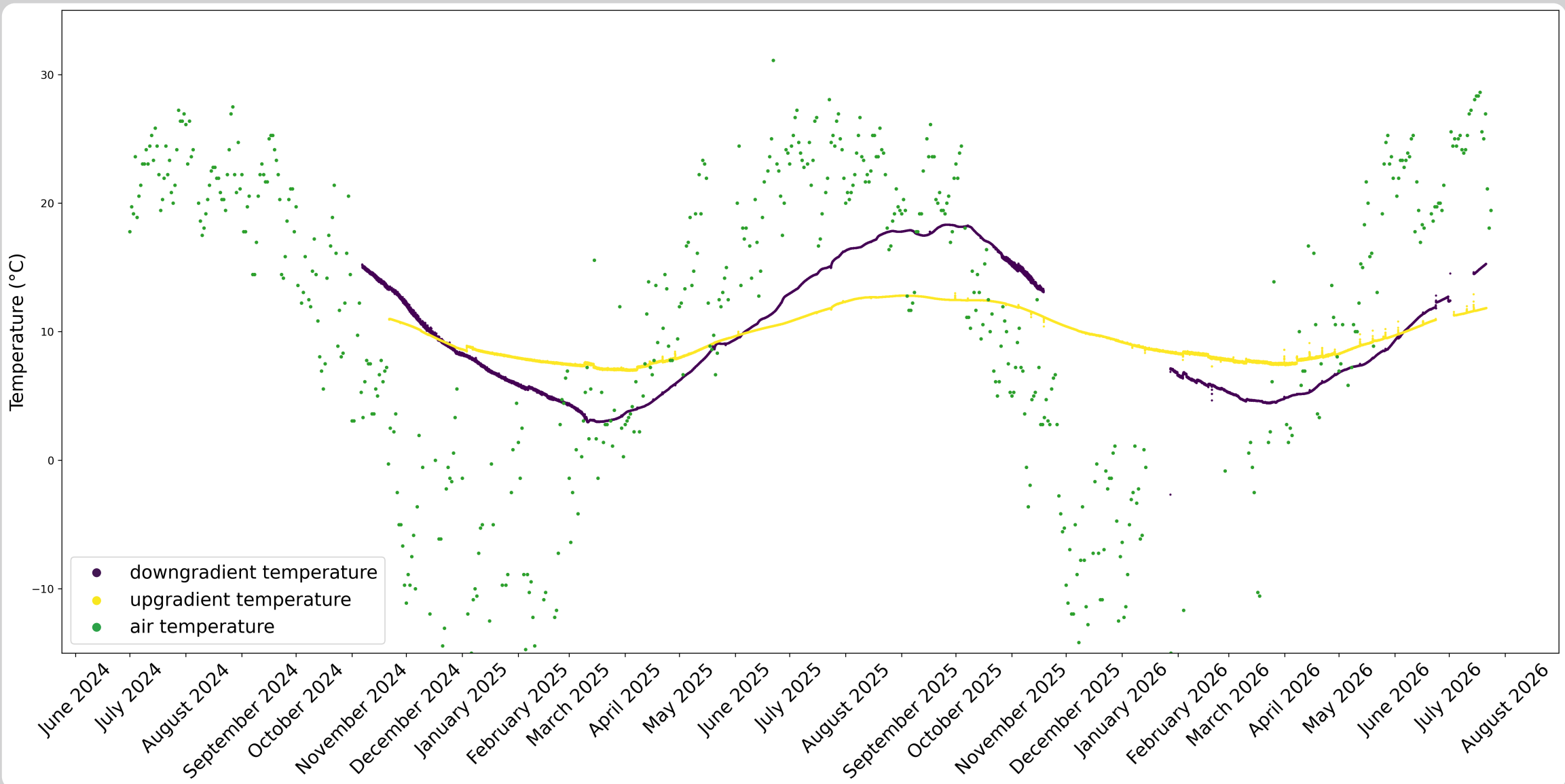
Adding monitoring wells will help us better understand groundwater flow beneath Bell Field.



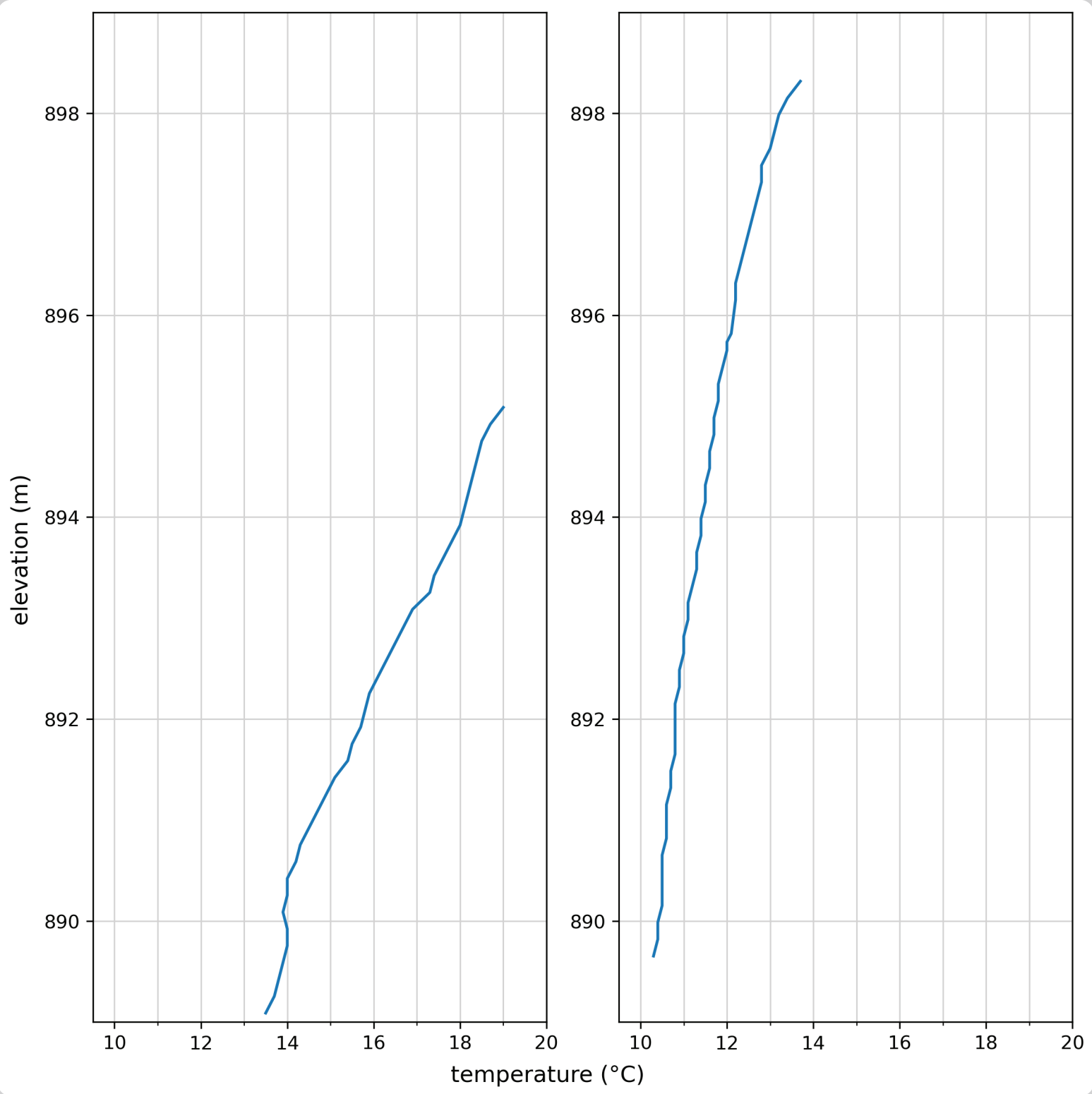
Results and main takeaways

Overall, we see evidence of a seasonal thermal plume but no evidence of it becoming more pronounced over time.

Temperature measurements across the monitoring wells reveal the possibility of the downgradient wells showing considerable impact from the geothermal system. Especially during summertime, when the cooling load is heaviest, we speculate that the borefield warms the shallow groundwater multiple degrees.

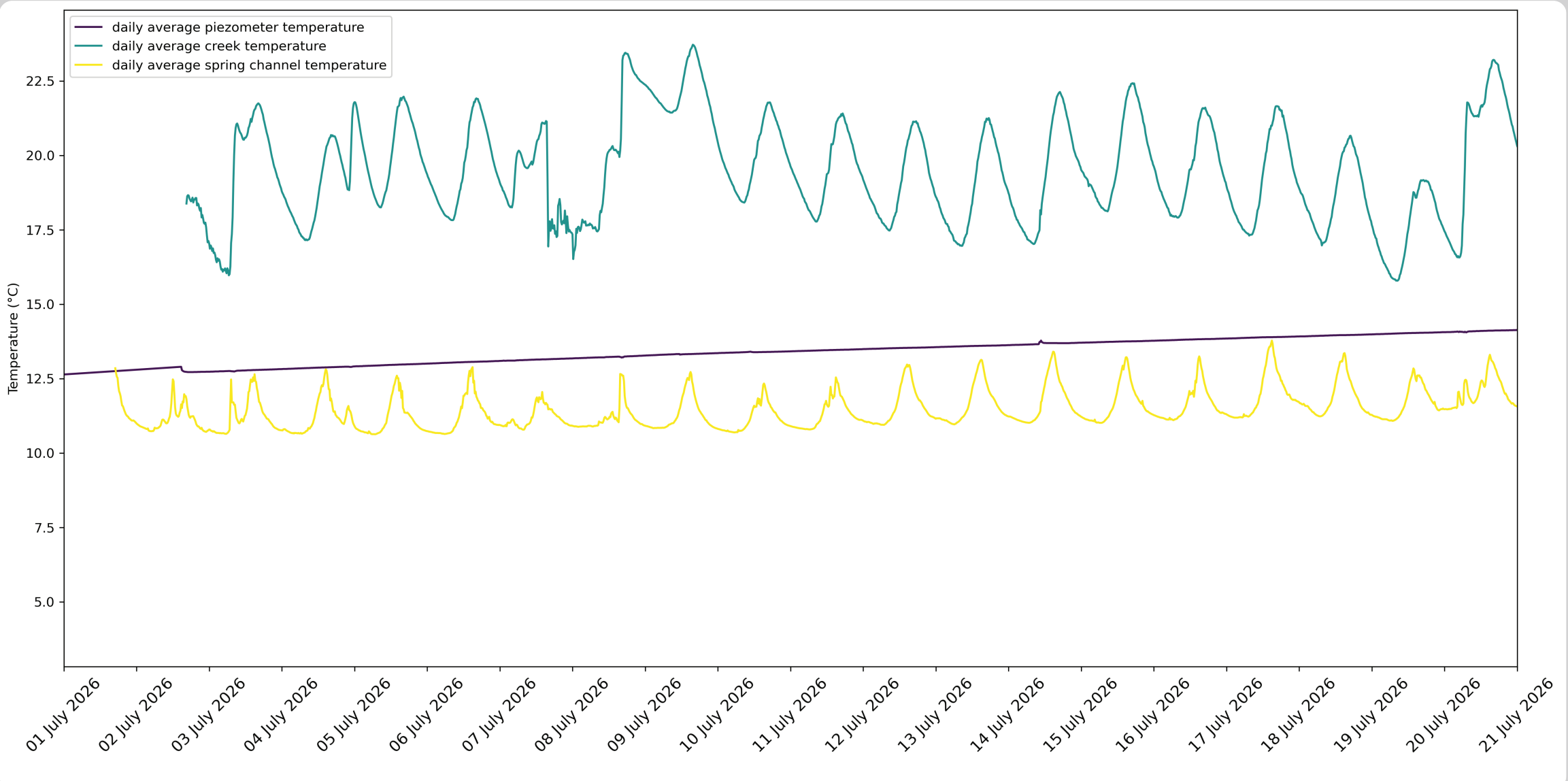


The monitoring well farthest upgradient stays closer to 10° C throughout the year, and the farthest downgradient well significantly warms in the summer and cools in the winter.



In late July, the upgradient piezometer is colder at the same elevations.

We suspect that the stream water temperature is buffered by groundwater flow from the opposite bank.



The spring channel is always colder than the water in the nearest piezometer.